XU ZHANG

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EDUCATION

M.Phil. in **Computer Science**, The University of Hong Kong *2021 - 2024*

B.Sc. in **Biology**, Shanxi Agricultural University *2013 - 2017*

- Rank: 3/30
- Graduation Thesis: 1/30

Visiting Student in **Computational Biology**, University of Georgia *2016 - 2017*

PROFESSIONAL EXPERIENCES

AI Bioinformatician *2024 - Present*

Research and Development Department, GeneNet Co. Ltd

Spatial Transcriptomics

- Built and maintained automated single-cell spatial transcriptomics pipelines using Spatial Ranger for raw data processing, Scanpy/Squidpy for clustering and biomarker identification, and Cell2Location for cell-type composition inference
- Integrated the pipeline into a commercial web application, streamlining data analysis workflows for end-users

Metabolomics

- Developed LC-MS/MS analysis tools using MetaboAnalystR for raw spectral processing, pathway enrichment, biomarker discovery, and statistical analyses
- Deployed containerized services on Cloud Run for efficient autoscaling, ensuring robust and scalable data processing solutions

Bioinformatics Platform Development

- Led a team to design and deploy a cloud-based analytics solution (Google Cloud, Next.js, Firebase) for multi-omics data
- Implemented RESTful APIs (Node.js, Express, FastAPI) to automate metabolic, transcriptomic, and genomic data analysis
- Received the Alibaba Cloud Silver Award and Excellence in AI Application for innovation and impact

MPhil Student *2021 - 2024*

Supervisors: Prof. HF Ting and Prof. TW Lam, The University of Hong Kong

Multi-Gene Prognostic Biomarkers

- Engineered a web application to identify potential multi-gene prognostic signatures by integrating a chatbot for addressing user queries, a module for identifying multi-gene prognostic biomarkers, and a module for automatically generating tailored reports
- Published as the first and corresponding author in *BIBM* and delivered an oral presentation

Genome Instability Associated Prognostic Signature for Lung Adenocarcinoma

- Developed a novel risk model by analyzing genomic, transcriptomic, and clinical data from 397 TCGA patients using univariate and multivariate Cox regression. Validated prognostic accuracy via Kaplan-Meier survival curves and log-rank testing.
- Published as the first author in *Frontiers in Cell and Developmental Biology*

Research Assistant *2019 - 2021*

Supervisor: Prof. Chu-Xia Deng, University of Macau

Genomics and Transcriptomics

- Led the genomic and transcriptomic data analysis of more than 100 breast cancer samples, including calling single-nucleotide variant (SNV) and copy number variation (CNV) from tumors as well as tumor-derived organoids, and identifying differentially expressed genes (DEGs) between drug-sensitive and drug-resistant groups of tumor-derived organoids to predict drug-resistant tumors
- Published as the second author in Advanced Science

Single-Cell Transcriptomics

- Led the analysis of scRNA-seq (10X) data to explore alterations in tumors and their microenvironment following PARPi resistance, focusing on identifying subclusters and biomarkers. Inferred intra-tumor heterogeneity and tumor evolution tree by analyzing single-cell genomic data
- Published a paper in *Theranostics*

Bioinformatics Engineer *2017 - 2019*

Research and Development Department, GENE Co. Ltd

Genomics

- Created and supported pipelines for high-throughput sequencing data analysis, accommodating both tumor-only and tumor–normal samples, and adapted these pipelines for cloud-based environments
- Enabled customers without programming expertise to efficiently process complex genomic data

Visiting Student *2016 - 2017*
Supervisor: Prof. Ying Xu, University of Georgia

Transcriptomics

- Mined TCGA's genomic and transcriptomic data, as well as visualized analysis results

PUBLICATIONS

Primary Author Publications:

- **Xu Zhang**, Lei Chen. MulMarker: a comprehensive framework for identifying multi-gene prognostic signatures. ***2023 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)***, IEEE Computer Society, 2023, **Oral Presentation**
- **Xu Zhang**, Tak-Wah Lam, and Hing-Fung Ting. Genome instability-derived genes as a novel prognostic signature for lung adenocarcinoma. ***Frontiers in Cell and Developmental Biology (IF: 6.08)***, 2023, 11
- Ping Chen, **Xu Zhang**, Renbo Ding, et al. Development of tailored therapy for advanced breast cancer based on patient-derived organoids and precision oncology. ***Advanced Science (IF: 17.52)***, 2021, 8(22)
- **Xu Zhang**, Baoyu Le, Zhimin Chen, et al. Establishment and Biological Characteristics Analysis of Fibroblast Cell Line of Guangling Donkey. ***China Animal Husbandry & Veterinary Medicine***, 2018, 45(3)

Contributing Author Publications:

- Jianjie Li, Xiaodong Shu, Jun Xu, Sek Man Su, Un In Chan, Lihua Mo, Jianlin Liu, Xin Zhang, Ragini Adhav, Kai Miao, Yuqing Wang, Tingting An, **Xu Zhang**, and NgaTeng IU, Chu-Xia Deng, Xiaoling Xu. S100A9-CXCL12 Axis in BRCA1 Deficiency Orchestrates the Immunosuppressive Microenvironment to Promote Breast Cancer Formation. ***Nature Communications (IF: 17.69)***, 2022, 13(1)
- Heng Sun, Jianming Zeng, Zengqiang Miao, Johnny Kuan Cheok Lei, Chen Huang, Lingling Hu, Sek Man Su, Un In Chan, Kai Miao, **Xu Zhang**, Aiping Zhang, Sen Guo, Ya Meng, Chris Koon Ho Wong, Xiaohua Douglas, Xiaoling Xu, Chu-Xia Deng. Dissecting the heterogeneity and tumorigenesis of BRCA1 deficient mammary tumors via single cell RNA sequencing. ***Theranostics (IF: 11.56)***, 2021, 11(20)
- Licen Li, Jiaolin Bao, Haitao Wang, Cheng Peng, Haipeng Lei, Jianming Zeng, Wenhui Hao, **Xu Zhang**, Xiaoling Xu, Chundong Yu, Chu-Xia Deng and Qiang Chen. Upregulation of amplified in breast cancer 1 contributes to pancreatic ductal adenocarcinoma progression and vulnerability to blockage of hedgehog activation. ***Theranostics (IF: 11.56)***, 2021, 11(4)
- Fangyuan Shao, Xueying Lyu, Kai Miao, Haitao Wang, Qiang Chen, Hao Xiao, Renbo Ding, Ping Chen, Fuqiang Xing, **Xu Zhang**, Guang-Hui Luo, Wenli Zhu, Li-Gong Lu, Gregory Cheng, Ng Wai Lon, Scott E. Martin, Guanyu Wang, Guokai Chen, Chu-Xia Deng. Enhanced protein damage clearance induces broad drug resistance in multi-type of cancers revealed by an evolution drug resistant model and genome-wide siRNA screening. ***Advanced Science (IF: 17.52)***, 2020, 7(23)
- Jianlin Liu, Ragini Adhav, Kai Miao, Sek Man Su, Lihua Mo, Un In Chan, Xin Zhang, Jun Xu, Jianjie Li, Xiaodong Shu, Jianming Zeng, **Xu Zhang**, Xueying Lyu; Lakhansing Pardeshi, Kaeling Tan, Heng Sun; Koon Ho Wong, Chu-Xia Deng; Xiaoling Xu. Characterization of BRCA1-deficient premalignant tissues and cancers by bulk and single-cell exome sequencing identifies Plekha5 as a tumor metastasis suppressor. ***Nature Communications (IF: 17.69)***, 2020, 11(1)
- Liang Sen, Sen Yang, Dayang Liang, Jiechao Ma, Yuan Tian, Jing Zhao, **Xu Zhang**, Ying Xu, and Yan Wang. A Novel Matched-pairs feature selection method considering with tumor purity for differential gene expression analyses. ***Mathematical Biosciences (IF: 3.935)***, 2019, 311

TEACHING

Teaching Assistant at COMP7409: Machine Learning in Trading and Finance (Class Size: 150) *2023*

- Led discussion sessions and office hours
- Supervised student projects, providing technical guidance on ML implementations

HONORS & AWARDS

Postgraduate Scholarships, The University of Hong Kong	*2021 - 2023*
Outstanding Graduates, Shanxi Agricultural University (top 10/800+)	*2017*
Academic Excellence Scholarship (thrice), Shanxi Agricultural University	*2014 - 2016*
Merit Student, Shanxi Agricultural University	*2015*
First Prize, the Fourth CP RBC Cup Competition of Shanxi Province	*2014*
Third prize (team leader), Shanxi Agricultural University Debating Competition	*2013*

ADDITIONAL INFORMATION

Review Experience: International Journal of Biological Sciences, Heliyon
Skills: Linux, R, Python, C/C++, JavaScript, TensorFlow, PyTorch, LangChain
Certifications: Google Cloud Computing Foundations Certificate, Google Cloud Data Analytics Certificate, Multi AI Agent Systems (CrewAI)
Languages: Chinese Mandarin (Native), English (Fluent)